

Evaluating the Brain Predictivity of Text Embeddings: A Comparison Between Large Language Models and Text Embedders

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Abstract

Brain encoding studies of language have largely relied on autoregressive, decoder-only large language models (LLMs), extracting passage-level representations via heuristics such as last-token or mean pooling. Dedicated text embedders, fine-tuned from LLM backbones with contrastive or instruction-following objectives, are designed to produce semantically dense passage-level representations, yet remain largely unevaluated as models of neural language processing. We compare Qwen3-Embedding against the matched decoder-only Qwen3 at three sizes (0.6B, 4B, 8B), predicting fMRI responses from the Pereira et al. (2018) dataset in the left-hemisphere language network. Across both experiments and all sizes, the two model families exhibit distinct layer-wise brain-predictivity profiles: the embedder peaks in early-to-mid layers while the LLM peaks in late layers, despite sharing the same causal attention architecture. A Toneva residual analysis further shows that the two architectures carry partially complementary brain-relevant information. These results suggest that training objective, rather than attention directionality, shapes how text-level semantic information is distributed across layers, and that dedicated embedders are competitive candidate models of passage-level neural language representations.

1 Introduction

Predicting brain activity provides a principled framework for evaluating computational models of language: rather than relying solely on downstream NLP benchmarks, it tests whether a model’s internal representations predict neural responses elicited during comprehension. In this framework, textual stimuli are embedded by a model and used to predict measured brain activity via voxel-wise regression, grounding model comparisons in measured neural activity. This study focuses specifi-

cally on *passage-level* semantic processing, asking which model representations best predict fMRI responses to multi-sentence passages rather than isolated words or sentences. Recent work demonstrates that representations from transformer language models account for substantial variance in human neural responses to language, particularly when those models are trained for next-word prediction (Schrimpf et al., 2021; Goldstein et al., 2022; Caucheteux and King, 2022). This has made large language models (LLMs) an increasingly important tool in cognitive neuroscience, where they serve as computational hypotheses about how linguistic information is structured and processed.

The majority of brain-alignment research has focused on autoregressive, decoder-only LLMs. When researchers extract passage-level representations from these models, they typically rely on heuristics such as isolating the final token’s hidden state or computing a mean-pooled average across all tokens (Tuckute et al., 2024; Hosseini et al., 2024). Such strategies are necessary because autoregressive models do not naturally output a single embedding for the full input. While these approaches yield statistically significant brain predictivity, recent NLP work suggests they may be inherently suboptimal for capturing global text meaning. The causal attention mechanism of decoder-only models conditions each token only on preceding context, prioritizing incremental prediction over holistic text representation (Zhang et al., 2025a; Su et al., 2025; Thennal D K et al., 2025). More recently, NLP research has produced dedicated text embedders, models initialized from LLM backbones but fine-tuned with contrastive or instruction-following objectives to optimize specifically for text representation (Muennighoff et al., 2023). Despite their widespread use in retrieval and semantic similarity tasks, these models remain largely unevaluated as candidate models of neural language representations.

This project addresses this gap by asking: are text embeddings from dedicated text embedders (the Qwen3-Embedding family) more brain-predictive than those from comparable decoder-only LLMs (the Qwen3 family), and does this hold across model scales? We compare matched pairs at three sizes (0.6B, 4B, 8B), which controls for model family and parameter count and isolates the effects of training objective from the effects of scale. Results are broadly consistent with this hypothesis. Across experiments conducted with stimuli and fMRI recordings, the text embedder achieves higher brain predictivity than the matched decoder-only LLM in the early layers of the network, while the LLM peaks in the final layers. Voxel-wise permutation testing at each model’s peak layer confirms that both architectures achieve above-chance predictivity (Section 4.7), and their layer-wise profiles replicate qualitatively across the two experiments. The unique-variance analysis further shows that the two architectures carry partially complementary brain-predictive information. The observed pattern is consistent with the idea that autoregressive models continue refining representations throughout the full network in service of next-token prediction, whereas embedding-specific training objectives concentrate text-level semantic information early in the processing stream.

Our main contributions are: (i) the first controlled comparison, to our knowledge, between a dedicated text embedder and a matched decoder-only LLM as encoding models for passage-level fMRI responses; (ii) evidence that the two architectures produce qualitatively distinct layer-wise brain-predictivity profiles that replicate across three model sizes and two independent experiments; and (iii) evidence from a Toneva residual analysis that the two architectures carry partially complementary brain-relevant information. Crucially, Qwen3-Embedding retains the causal attention architecture of its Qwen3 backbone, so this comparison isolates the effect of the embedding-specific training objective rather than a difference in attention directionality.

2 Related Work

2.1 Brain Encoding Models for Language

A central question in brain predictivity is which computational representations best predict neural responses during language comprehension. Early work relied on static lexical or distributional repre-

sentations, such as Mikolov et al. (2013) and Pennington et al. (2014), but the advent of deep transformer architectures marked a substantial shift.

Subsequent work strengthens the case for transformer language models as computational models of the language system. Schrimpf et al. (2021) compare a broad set of language transformer models against neural and behavioral benchmarks, reporting that next-word prediction performance is strongly associated with brain alignment, a finding that suggests predictive processing may be a shared computational principle between human cognition and artificial networks (Goldstein et al., 2022). At a finer level of analysis, Caucheteux and King (2022) show that transformer representations exhibit meaningful layer-wise correspondences with brain activity. This layer-wise perspective is especially relevant for studies comparing representational dynamics across architectures. In parallel, Antonello et al. (2023) demonstrate that encoding performance scales with model size, but that informative representational structure is already present in relatively small models, making them suitable and computationally efficient for voxel-wise encoding analyses. Together, these studies establish brain encoding as a productive comparative framework and motivate layer-wise analyses rather than evaluations based solely on final-layer representations.

2.2 Autoregressive LLMs vs. Embedding-Specialized Models

Although autoregressive LLMs dominate recent brain-alignment work, they are not the only plausible source of brain-relevant text representations. The distinction between general-purpose LLMs and embedding-specialized models matters because the two are optimized under different objectives. Decoder-only models condition each token on left context and are optimized for continuation, which makes them natural candidates for modeling predictive aspects of language processing (Schrimpf et al., 2021; Goldstein et al., 2022). However, the same next-token objective that supports prediction may also make sequence-level representations less direct: global meaning is not encoded under an objective that explicitly encourages a single, semantically stable sequence-level vector.

Earlier work at the intersection of NLP and cognitive neuroscience suggests that training-objective differences affect neural alignment. Toneva and Wehbe (2019) compare several contextual models, including BERT (Devlin et al., 2018) and

Transformer-XL (Dai et al., 2019), and show that layer depth, context length, and attention pattern all influence how model representations relate to brain data. Gauthier and Levy (2019), working in a sentence-decoding setting, find that fine-tuning BERT on additional Natural Language Understanding (NLU) tasks does not automatically improve correspondence with neural representations, implying that not all task-optimized sentence representations are equally brain-relevant. These results do not establish a simple superiority of either model family, but they do suggest that training objective and representational geometry matter for neural fit, making a controlled comparison between a decoder-only LLM and a matched text embedder particularly informative.

2.3 Dedicated Text Embedders

A recent development in NLP is the emergence of models designed specifically to produce high-quality text embeddings, rather than being repurposed from base LLMs. These text embedders are typically initialized from a pre-trained LLM backbone and undergo extensive secondary training with contrastive learning or instruction-following objectives (Su et al., 2025; Zhang et al., 2025a; Thennal D K et al., 2025). Importantly, many modern text embedders retain the causal attention mechanism of their decoder-only backbone; the key difference lies in the training objective (contrastive or retrieval-oriented) rather than the attention pattern. The MTEB benchmark formalizes this landscape by evaluating embedding models across a broad range of tasks and languages, demonstrating that embedding quality depends not only on base-model scale but also on training objective and model specialization (Muennighoff et al., 2023). However, strong performance on NLP benchmarks measuring semantic similarity or retrieval does not automatically guarantee alignment with human neural processing.

Qwen3-Embedding, the model evaluated in the present study, is a member of the Qwen3 family optimized specifically for embedding tasks rather than general autoregressive generation (Zhang et al., 2025b). It shares the causal attention architecture of its decoder-only counterpart but is fine-tuned with contrastive and instruction-following objectives to produce semantically dense representations. The Qwen3 family makes it possible to compare the embedder against a matched-size decoder-only backbone (Yang et al., 2025). This distinction mat-

ters because a base LLM is optimized to predict the next token; any useful text representation is therefore a by-product of that objective rather than its primary target. A dedicated embedder instead treats semantically informative text compression as the primary target, potentially producing representations that are more compact, more globally semantic, and more stable across layers.

From a brain-encoding perspective, this raises a clear empirical question: if the neural target reflects integrated passage meaning rather than incremental next-word prediction, do embedding-specialized models provide a better representational match? Existing brain-alignment literature has not directly tested this question with a controlled comparison between a dedicated embedder and a matched decoder-only backbone. The present study is motivated by that gap.

3 Methods

3.1 Data

This study uses the fMRI dataset collected by Pereira et al. (2018), who conducted three experiments on a total of 16 participants. During these experiments, participants were presented with linguistic stimuli while their brain activity was recorded via fMRI. Since Experiment 1 involved pictorial stimuli, this study focuses exclusively on Experiments 2 and 3.

Experiment 2 was conducted on 8 participants and contained 96 passages, each consisting of 4 sentences about one of 24 topics (e.g., professions, musical instruments), with 4 passages per topic. The 384 sentences range from 7 to 18 words in length (mean = 11.8, SD = 2.1). All passages were Wikipedia-style texts providing factual information about the relevant concept.

Experiment 3 was conducted on 6 participants and contained 72 passages consisting of 3 or 4 sentences about 24 topics that did not overlap with those from Experiment 2. There were 3 passages per topic, comprising 48 Wikipedia-style passages and 24 first- or third-person narratives.

Five participants completed both experiments, but Experiments 2 and 3 are analyzed independently. Brain encoding models were trained on both passage-level and sentence-level embeddings.

3.1.1 Voxel Representation

The fMRI responses are represented as vectors of voxel-wise activations, where each voxel cor-

Model	Parameters	Layers	Hidden dim.
Qwen3-0.6B	0.6B	28	1024
Qwen3-4B	4B	36	2560
Qwen3-8B	8.2B	36	4096

Table 1: Hyperparameters of the models used in the experiments. The embedding versions share the same architecture.

responds to a unit of 3D brain volume. Since this study concerns language processing, the analysis is restricted to voxels within the functionally localized left-hemisphere language network, which serves as the Region of Interest (ROI). The number of voxels in this ROI varies across participants due to individual differences in functional localization.

3.2 Embedding Models

The models used for embedding extraction originate from the Qwen3 family. From this family we compare Qwen3-0.6B, Qwen3-4B, and Qwen3-8B, as well as their respective dedicated embedding models. We examine whether larger models yield higher brain predictivity, given their greater number of parameters, layers, and embedding dimensions. An overview of these differences is shown in Table 1.

For each model, embeddings are extracted at every layer using two pooling strategies. For the decoder-only LLMs, the primary extraction method takes the hidden state of the last token at each layer; a mean-pooling variant that averages hidden states across all input tokens is also computed. For the text embedders, an EOS token is appended to the input, and the hidden state corresponding to this token at each layer serves as the primary embedding (matching the model’s intended readout); a mean-pooling variant is likewise extracted. All four pooling/architecture combinations are compared, enabling a systematic evaluation of how both model type and extraction strategy affect brain predictivity. Randomly initialized (untrained) versions of both architectures serve as baselines.

These models were selected because they form controlled pairs: matched in size and architecture family, but differing in whether they were optimized for autoregressive generation or for text embedding.

3.3 Encoding Model

Brain predictivity is assessed using voxel-wise encoding models (Naselaris et al., 2011). For each

voxel, a separate kernel ridge regression (Kernel-RidgeCV) is trained to predict the neural response from the model embedding. The regularization parameter α is selected from a range of 10^0 to 10^8 (20 values on a log scale) via an inner 5-fold cross-validation loop. Prediction accuracy is evaluated using an outer 5-fold cross-validation, computing Pearson correlation $r(\hat{y}, y)$ between predicted and observed voxel activations per fold, then averaging across folds and voxels.

This procedure is repeated independently for each model layer, yielding a layer-wise brain predictivity curve for each model. The same pipeline is applied to the randomly initialized baselines.

4 Results

4.1 Layer-Wise Brain Predictivity (Passages)

Figures 1 and 2 show the participant-averaged layer-wise brain predictivity for passage-level fMRI responses, faceted by model size (0.6B, 4B, 8B). Per-participant breakdowns are provided in Appendix A.

The two model families exhibit distinct layer-wise trajectories. The text embedder (Qwen3-Embedding, EOS pooling) shows its highest brain predictivity in the early-to-mid layers of the network, while the decoder-only LLM (Qwen3, last-token pooling) starts near $r \approx 0$ and rises through the network, reaching its highest predictivity in the later layers. For the 0.6B models, the embedder’s early-layer predictivity reaches $r \approx 0.18$ -0.20, while the LLM peaks at $r \approx 0.15$ -0.17. Randomly initialized baselines for both architectures remain near zero across all layers, confirming that the observed predictivity reflects learned representations rather than architectural inductive biases.

Both models’ layer-wise curves can contain multiple near-tied local maxima, so that the global argmax (used for permutation testing in Section 4.7) does not always coincide with the visually prominent peak. This is most apparent for the embedder, but also affects the LLM at 8B in Experiment 3 (Table 2: argmax at L=4 despite a visually late peak). The overall shape of the curve, rather than the single argmax layer, is therefore the more informative comparison.

This pattern replicates across all three model sizes. At 4B and 8B, both models’ peaks shift to correspondingly higher absolute layer indices, but their relative positions within each network remain consistent. The cross-size replication suggests that

the divergent trajectories reflect a difference in how the two training objectives distribute semantic information across layers, rather than an artifact of any single model configuration.

4.2 Comparison to Noise Ceiling

To contextualize the absolute predictivity values, Figures 3 and 4 overlay the layer-wise curves with the between-participant noise ceiling. The lower bound of the ceiling is computed as the leave-one-out correlation (each participant’s ROI-mean response correlated with the mean of the remaining participants), and the upper bound includes the target participant in the group mean.

The mean noise-ceiling bands are [0.17, 0.49] for Experiment 3 passages and [0.14, 0.41] for Experiment 2 passages. The best-performing trained models reach the lower bound of the ceiling ($r \approx 0.17$ -0.20), indicating that they explain roughly as much variance as the consistency between one participant’s ROI-mean response and the leave-one-out group average. However, all models remain well below the upper bound, suggesting substantial room for improvement.

4.3 Sentence-Level Results

The same qualitative pattern holds at the sentence level (Figures ?? and ??): the embedder shows higher predictivity in early layers and the LLM peaks late. However, absolute peak correlations are somewhat lower than at the passage level (e.g., embedder peaks at $r \approx 0.16$ -0.17 for Experiment 3 sentences, compared to $r \approx 0.18$ -0.20 for passages), consistent with the expectation that sentence-level fMRI responses are noisier due to less signal averaging per condition.

4.4 Ablations and Follow-up Analyses: Pooling and Model Size

Mean-pooling variants of both architectures generally produce brain-predictivity curves that are flatter and more moderate ($r \approx 0.13$ -0.18) than the primary pooling variants, though some mean-pool configurations peak at unexpected layers (e.g., early layers for certain larger models). This suggests that mean-pooling dilutes the layer-specific information that each architecture concentrates at its respective peak under its primary pooling strategy.

Across model sizes, larger models (4B, 8B) do not substantially outperform the 0.6B model in peak brain predictivity, though they maintain

the same qualitative layer-wise pattern. The permutation-derived %FDR values (Tables 2 and 3) show a slight downward trend for the LLM last-token variant at larger sizes, suggesting that additional parameters do not straightforwardly translate to more voxels reaching significance. This is consistent with prior findings that brain predictivity scales sub-linearly with model size (Antonello et al., 2023).

4.5 Intrinsic Dimensionality

To investigate what representational property underlies the observed brain-predictivity differences, we compute the participation ratio (a linear measure of intrinsic dimensionality) at each layer for the trained models (Figures 5 and 6).

The LLM’s last-token representations exhibit very low participation ratio (~ 1 -5) in the earliest layers, then climb to ~ 15 -50 in later layers. The embedder’s EOS representations start at moderate participation ratio (~ 10 -13) and remain relatively stable or decline slightly through the network. Randomly initialized baselines show flat, high participation ratio (> 30) from layer 0, consistent with unstructured noise.

Figures ?? and ?? plot mean test r against participation ratio across all layers and models. The relationship is not a simple monotonic trend: the embedder achieves its highest brain predictivity at moderate participation ratio in the early layers, while the LLM starts with very low participation ratio but also low brain predictivity. The random-init baselines cluster at high participation ratio and near-zero brain predictivity. This suggests that neither extremely low nor extremely high intrinsic dimensionality is optimal for brain predictivity; rather, the combination of training objective and representational compression jointly determines neural fit.

4.6 Unique Variance (Toneva 2022)

To assess whether the two architectures capture complementary brain-relevant information, we apply the residual method of Toneva et al. (2022). For each model size, the unique variance that one model’s peak-layer embeddings explain beyond the other is estimated in both directions using a two-stage KernelRidgeCV procedure (Figures 7 and 8). Bar labels indicate the peak layer chosen for each model in the format L{target}|L{other}.

Both architectures carry substantial unique brain-predictive information beyond what the other cap-

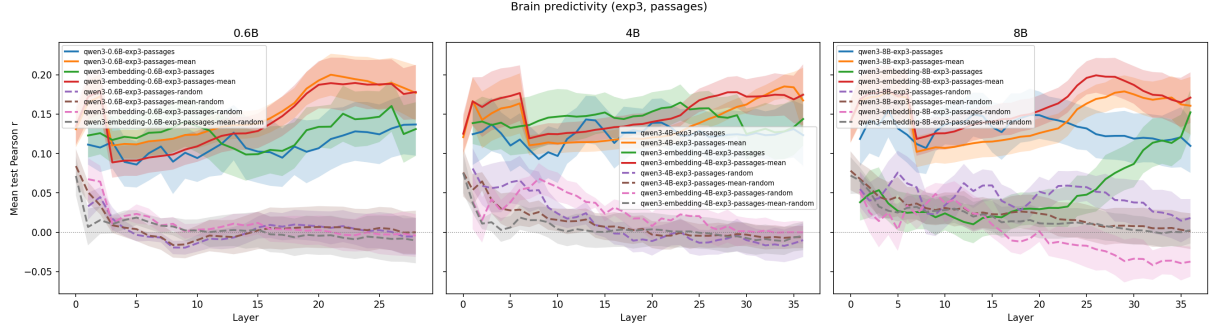


Figure 1: Participant-averaged layer-wise brain predictivity for Experiment 3 (passages, $N = 6$), faceted by model size. Solid lines: trained models; dashed lines: randomly initialized baselines. Shaded regions: standard error across participants.

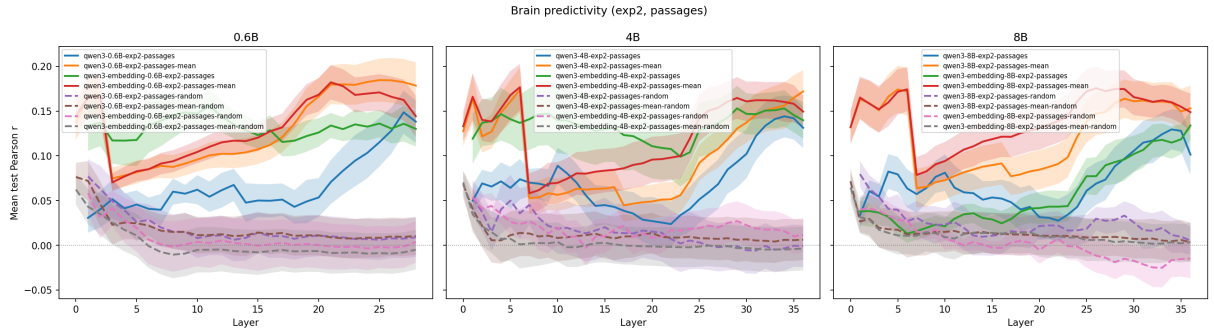


Figure 2: Participant-averaged layer-wise brain predictivity for Experiment 2 (passages, $N = 8$), faceted by model size.

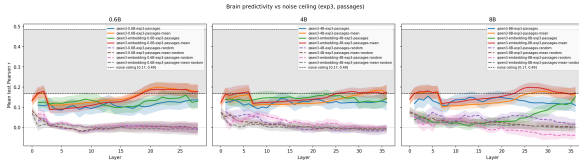


Figure 3: Experiment 3 passage-level brain predictivity with noise-ceiling overlay (shaded band: lower and upper bound).

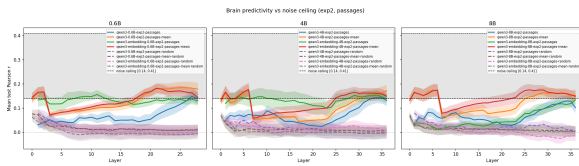


Figure 4: Experiment 2 passage-level brain predictivity with noise-ceiling overlay.

tures ($r \approx 0.10$ - 0.15 in unique variance in both directions). At the 0.6B and 4B sizes, the asymmetry slightly favors the embedder: it contributes more unique variance above the LLM than vice versa. At 8B, the asymmetry shrinks and the two architectures become roughly balanced or the LLM marginally exceeds. This pattern suggests that while the two model families share a substan-

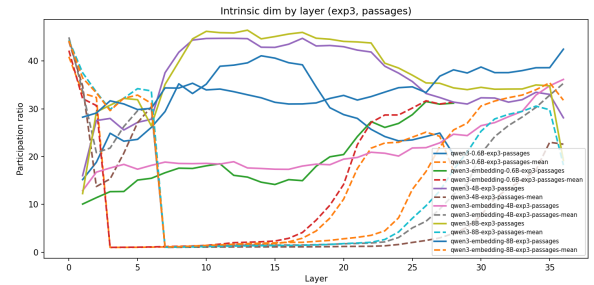


Figure 5: Participation ratio (intrinsic dimensionality) per layer for Experiment 3 passages.

tial representational core, each also encodes brain-relevant information that the other does not, and that this complementarity is most pronounced at smaller model sizes.

4.7 Statistical Significance

To verify that the observed brain predictivity exceeds chance, we conduct a canonical refit permutation test at each model's peak layer (passages only, where peak layer is defined as the argmax of the participant-averaged mean test r). For each of 1000 permutations the stimulus-to-brain mapping is shuffled and the full encoding pipeline is refit,

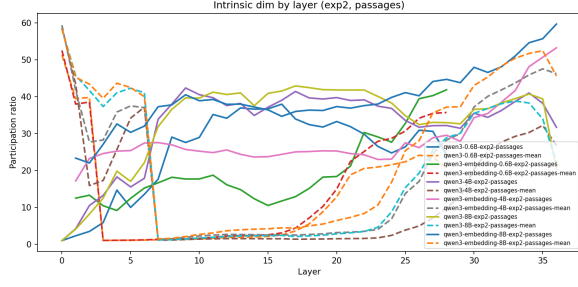


Figure 6: Participation ratio per layer for Experiment 2 passages.

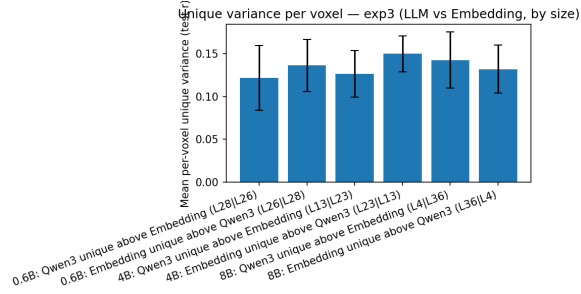


Figure 7: Unique variance (Toneva residual) for Experiment 3 passages. Each pair of bars shows the unique brain-predictive variance of one model above the other, at each model size.

yielding a per-voxel null distribution of Pearson correlations. Significance is assessed per voxel under both Benjamini-Hochberg FDR ($q < 0.05$) and Bonferroni correction.

Tables 2 and 3 report the resulting per-model peak layers and the percentage of voxels surviving each correction, averaged across participants. Note that the argmax-selected peak layer can differ from the visually prominent peak in the averaged curves when the curve contains multiple near-tied local maxima (see Section 4.1).

Under FDR correction, between $\sim 9\%$ and $\sim 28\%$ of voxels reach significance per model. The highest %FDR values are obtained for the mean-pool variants (e.g., Qwen3-mean at 27.9% for Experiment 2 0.6B; Qwen3-Embed-mean at 24.9% for Experiment 3 8B), reflecting both their broader layer-wise plateau and the higher per-fit stability that mean pooling provides. The last-token and EOS variants reach somewhat lower peak-layer %FDR values ($\sim 10\text{-}16\%$) but, as Sections 4.1 and 4.2 show, achieve comparable or higher absolute predictivity at their visually prominent peaks.

The Bonferroni column is uniformly 0% at the 1000-permutation resolution used here (see table captions for details). Full-resolution voxel-wise

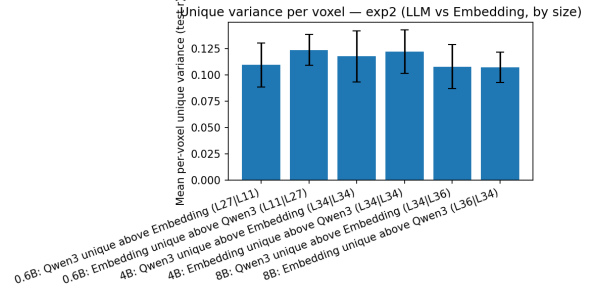


Figure 8: Unique variance (Toneva residual) for Experiment 2 passages.

Model	Peak L	%FDR	%Bonf
0.6B			
Qwen3 (last)	28	16.0	0.0
Qwen3 (mean)	21	24.3	0.0
Qwen3-Embed (EOS)	26	14.9	0.0
Qwen3-Embed (mean)	23	20.3	0.0
4B			
Qwen3 (last)	13	11.7	0.0
Qwen3 (mean)	34	18.0	0.0
Qwen3-Embed (EOS)	23	17.1	0.0
Qwen3-Embed (mean)	29	19.0	0.0
8B			
Qwen3 (last)	4	15.9	0.0
Qwen3 (mean)	29	19.5	0.0
Qwen3-Embed (EOS)	36	11.7	0.0
Qwen3-Embed (mean)	26	24.9	0.0

Table 2: Peak-layer voxel-wise significance for Experiment 3 passages ($N = 6$). %FDR = percent of voxels surviving BH-FDR at $q < 0.05$; %Bonf = percent surviving Bonferroni at $p < 0.05/n_{\text{voxels}}$, both averaged across participants. Bonferroni is uniformly 0% because 1000 permutations with add-one smoothing yield a minimum per-voxel $p \approx 10^{-3}$, above the Bonferroni threshold of $\approx 5 \times 10^{-6}$ for $\sim 10^3\text{--}10^4$ voxels.

Bonferroni correction would require $\geq 10^5$ permutations, which was beyond the available compute budget. The random-init baselines (Figures 1 and 2) sit near zero across all layers and were therefore not formally tested.

5 Discussion and Conclusion

This study compared dedicated text embedders against matched decoder-only LLMs as encoding models for passage-level fMRI responses in the left-hemisphere language network. Three main findings emerge.

First, the two model families exhibit distinct layer-wise brain-predictivity profiles. The embedder shows its highest predictivity in the early layers of the network, consistent with a training objective that targets semantically dense output rep-

Model	Peak L	%FDR	%Bonf
<i>0.6B</i>			
Qwen3 (last)	27	14.7	0.0
Qwen3 (mean)	25	27.9	0.0
Qwen3-Embed (EOS)	11	16.3	0.0
Qwen3-Embed (mean)	21	22.7	0.0
<i>4B</i>			
Qwen3 (last)	34	14.7	0.0
Qwen3 (mean)	6	19.5	0.0
Qwen3-Embed (EOS)	34	11.5	0.0
Qwen3-Embed (mean)	6	21.2	0.0
<i>8B</i>			
Qwen3 (last)	34	9.8	0.0
Qwen3 (mean)	5	20.3	0.0
Qwen3-Embed (EOS)	36	9.3	0.0
Qwen3-Embed (mean)	26	23.1	0.0

Table 3: Peak-layer voxel-wise significance for Experiment 2 passages ($N = 8$). Conventions as in Table 2.

representations. The LLM accumulates predictive power gradually across the full network, consistent with the incremental refinement demanded by next-token prediction. This contrast replicates across three model sizes and two independent experiments, suggesting it reflects a stable property of the respective training objectives rather than an artifact of any single configuration. Importantly, both models share the same causal attention architecture; the divergence in layer-wise profiles is therefore attributable to the training objective, not to differences in attention directionality.

Second, intrinsic dimensionality analysis reveals that the two model families occupy different regions of the participation-ratio/brain-predictivity space. The LLM’s last-token representations are highly compressed (low participation ratio) in the earliest layers but achieve low brain predictivity there, while the embedder starts at moderate participation ratio with high brain predictivity. This suggests that representational compression alone does not determine brain predictivity; rather, the combination of compression and training objective jointly shapes how well a representation predicts neural responses.

Third, the Toneva residual analysis shows that the two architectures carry partially non-overlapping brain-relevant information. This complementarity is most pronounced at smaller model sizes and diminishes at 8B, where the LLM’s richer parameterization may allow it to recover some of the information that the embedder captures through training-objective specialization.

These findings both extend and complicate prior

work on brain encoding with transformer language models. The observation that decoder-only LLM predictivity rises through the network is consistent with [Caucheteux and King \(2022\)](#) and [Schrimpf et al. \(2021\)](#), who report that mid-to-late transformer layers typically provide the best brain alignment. However, the embedder’s early-layer peak departs from this pattern and suggests that the “deeper is better” generalization is contingent on the training objective rather than being a general property of transformer architectures. The sub-linear scaling of peak predictivity with model size echoes [Antonello et al. \(2023\)](#), who similarly find that informative representational structure is already present in relatively small models. Finally, the partial complementarity revealed by the Toneva residual analysis aligns with [Toneva et al. \(2022\)](#), who originally proposed the method to show that different model components capture distinct aspects of meaning composition; here we extend that logic to a comparison between training objectives within a fixed model family.

These results have practical implications for brain-encoding studies. Researchers seeking to maximize brain predictivity from a single model should consider dedicated text embedders rather than defaulting to decoder-only LLMs, particularly when the neural target is a passage-level semantic representation. Furthermore, for passage-level encoding with dedicated text embedders, the early layers of the network may contain more brain-relevant information than the final layer, though the presence of multiple near-tied local maxima in the layer-wise curves (see Section 4.1) warrants caution in interpreting any single peak layer.

6 Limitations

Several limitations constrain the generalizability of these findings. First, statistical significance was assessed only at each model’s argmax peak layer via a canonical refit permutation test; a full layer-by-layer significance analysis would provide a more complete picture but was computationally prohibitive. The argmax-selected peak layer does not always coincide with the visually prominent peak due to near-tied local maxima, which means the permutation test may not be evaluating the layer where the model is most informative. Second, the Toneva residual analysis was restricted to the primary pooling strategy for each architecture (last-token for the LLM, EOS for the embedder);

intra-pooling comparisons (e.g., LLM last-token vs. LLM mean-pool) were not conducted. Third, all analyses targeted a single ROI, the functionally localized left-hemisphere language network, and it remains open whether the same patterns hold in other brain regions or in whole-brain analyses. Fourth, the two experiments were analyzed independently; no joint statistical test was performed across experiments, so the replication is informal. Finally, the models compared are all members of the Qwen3 family; whether the observed layer-wise divergence generalizes to other LLM/embedder pairs (e.g., Mistral vs. Mistral-Embed) remains to be tested.

Author Contributions

Ali Bilge led the fMRI data pipeline and preprocessing, implemented the noise-ceiling computation, intrinsic dimensionality analysis, and permutation-based significance testing, and wrote the Introduction, Related Work, Discussion, and Limitations sections. Jip de Vries led the embedding extraction pipeline across all model sizes and pooling variants, implemented the Toneva residual analysis, and wrote the Methods and Results sections. Both authors contributed to experimental design, figure generation, and editing of the final report.

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A Per-Participant Results

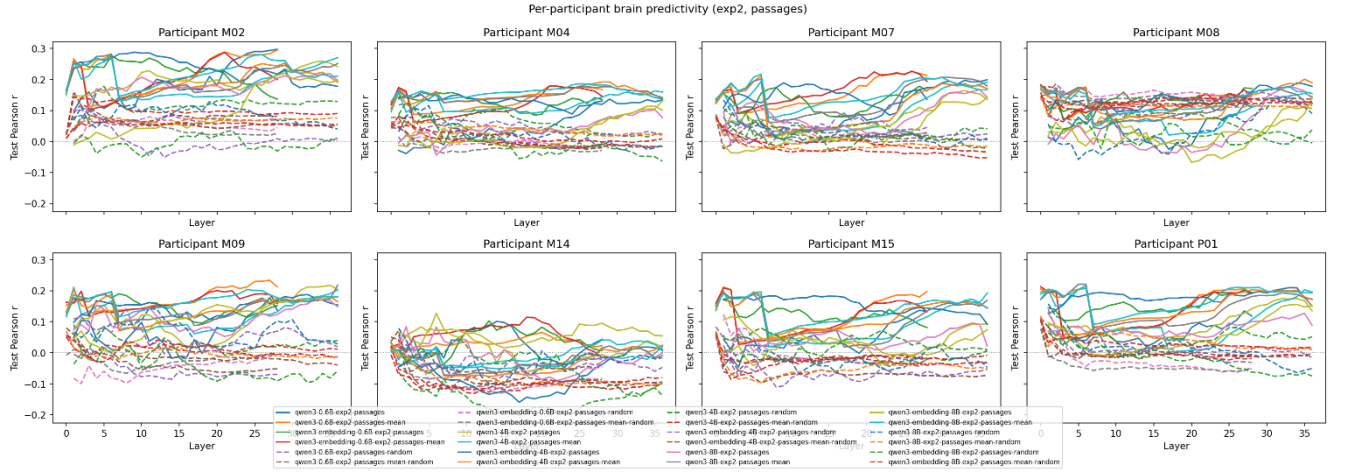


Figure 9: Layer-wise brain predictivity for Experiment 2 (passages, $N = 8$), shown per participant. Solid lines: trained models; dashed lines: randomly initialized baselines. size

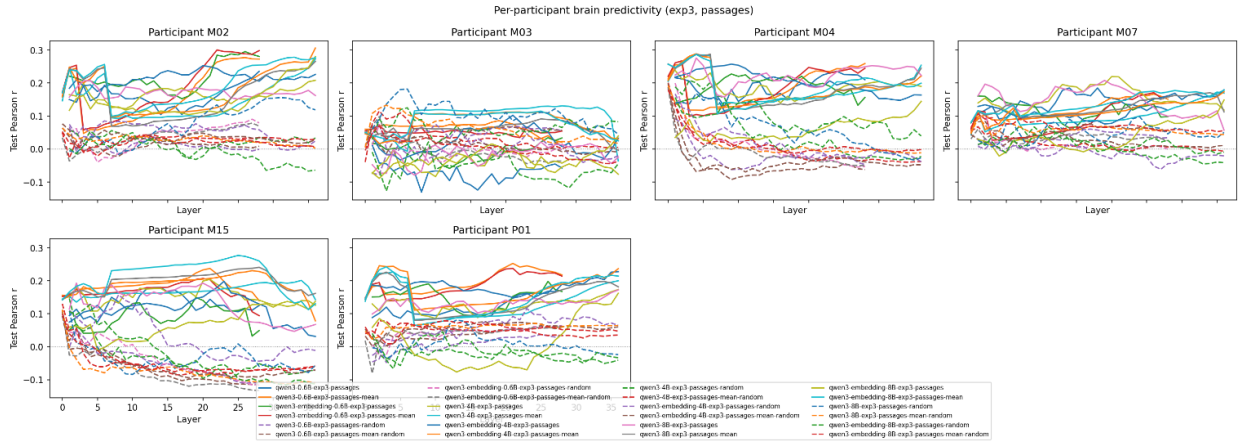
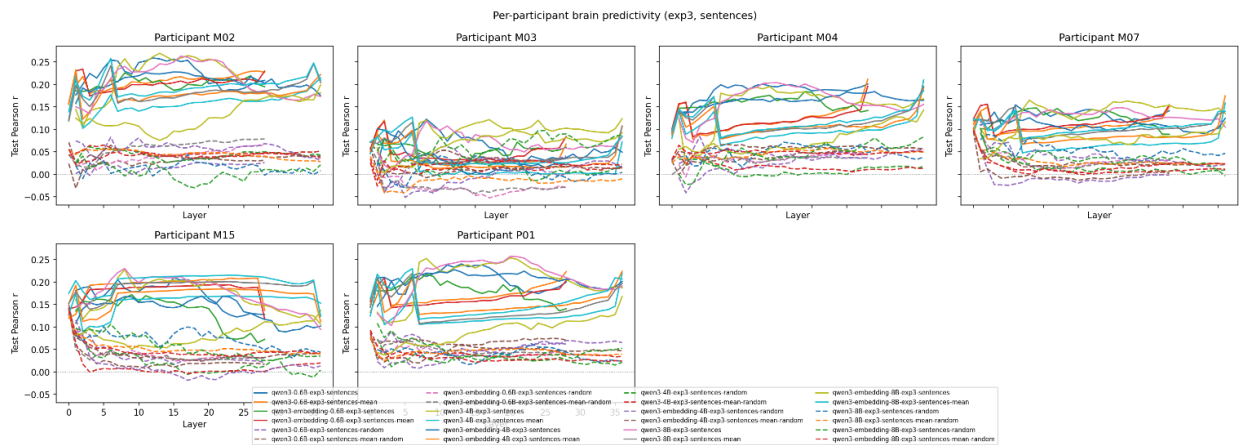
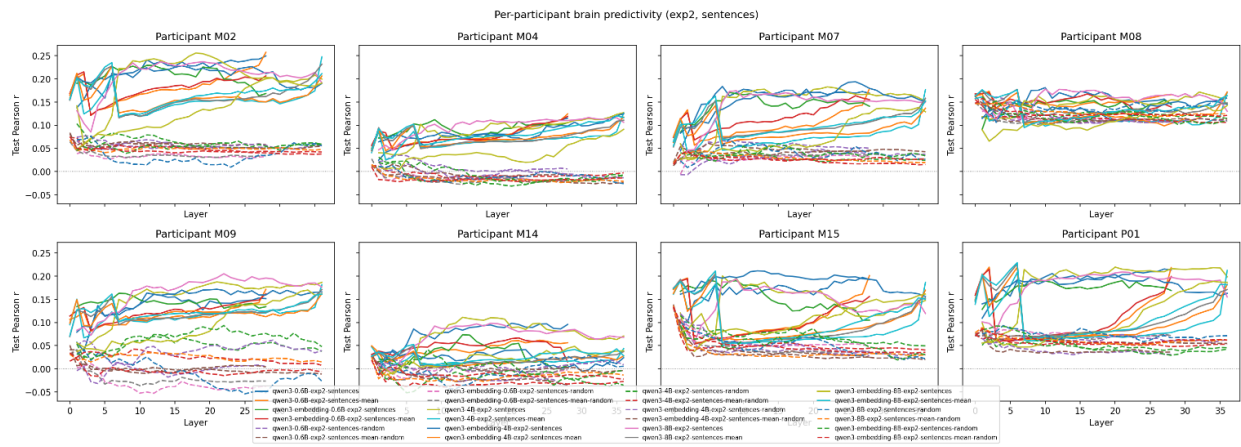


Figure 10: Layer-wise brain predictivity for Experiment 3 (passages, $N = 8$), shown per participant.



B Sentence-level Results

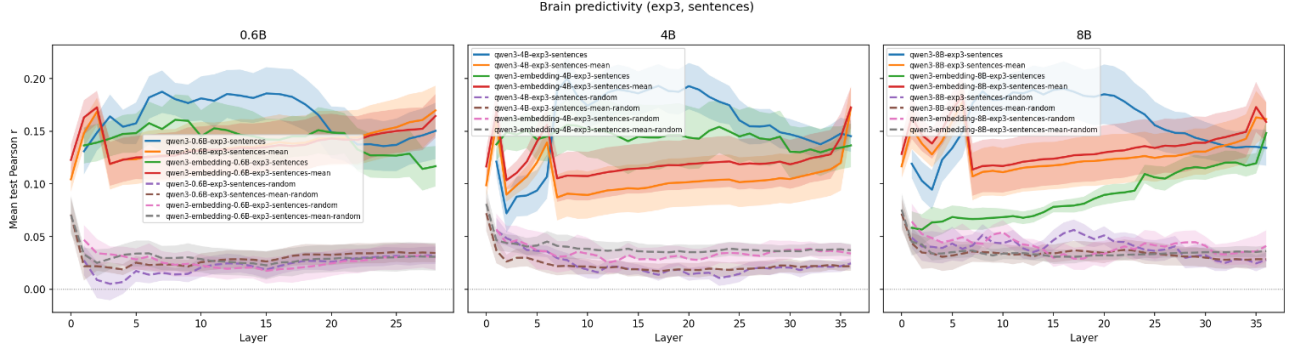


Figure 13: Experiment 3 sentence-level brain predictivity, faceted by model size.

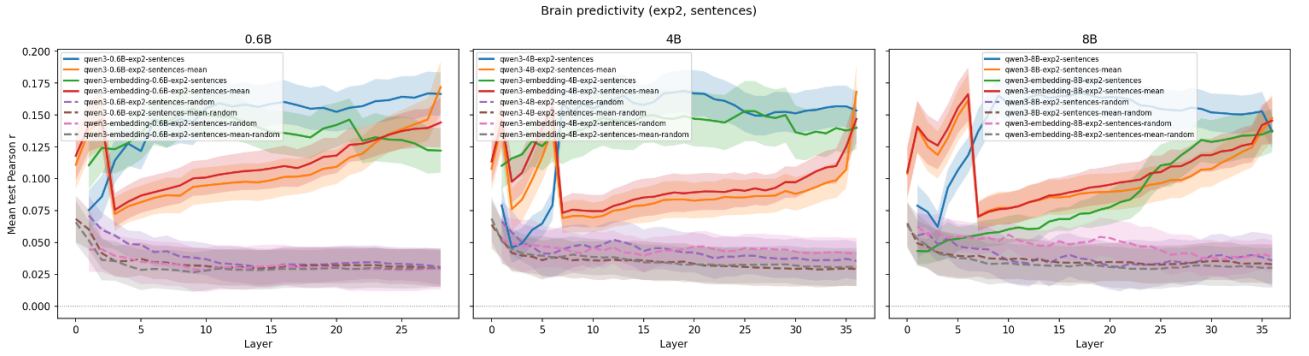


Figure 14: Experiment 2 sentence-level brain predictivity, faceted by model size.

C Intrinsic Dimensionality Scatter Plots

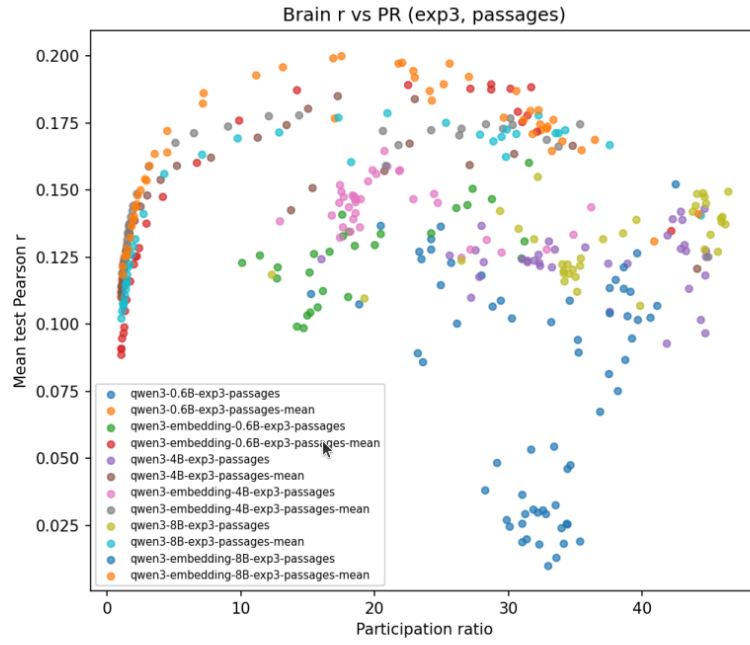


Figure 15: Participation ratio vs. mean brain predictivity (r) for Experiment 3 passages.

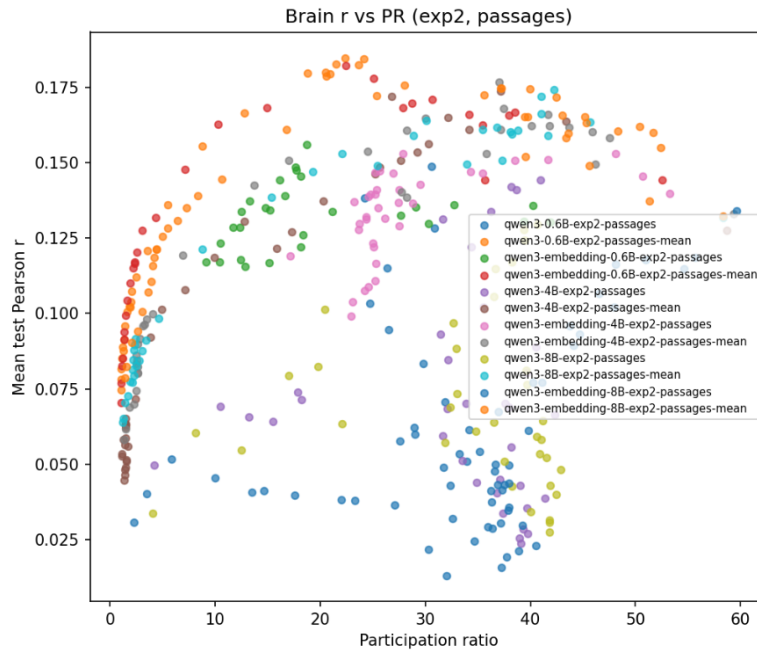


Figure 16: Participation ratio vs. mean brain predictivity (r) for Experiment 2 passages.